Department of Artificial Intelligence and Machine Learning

Course Code : **AI372IA** Course : **Stream Processing and Analytics**

Literature Survey

Project Title	Smart Agriculture Crop Health Monitoring
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1. Introduction

This literature review covers **30 papers** on Kafka and Flink stream processing, edge-fog- cloud architectures, streaming anomaly detection and agentic AI in precision agriculture. It identifies where existing systems fall short on false alerts, fault tolerance and verified recommendations, which are the gaps our project addresses. For each paper we note its key contribution and the research gap.

Literature Review

#	Authors, paper, venue and DOI	Key contribution	Research gap
1	**Agentic LLM-Driven Microservices over Event-Driven Kafka Bus for Plant Pathogen Early Warning** **In:** Frontiers in Plant Science, 2026.	Couples low-latency Kafka message brokers with specialized agentic LLMs configured as microservices to interpret continuous visual and environmental telemetry streams.	Opaque black-box ML outputs lacking actionable explanations for farm managers, and delayed fusion of unstructured textual knowledge bases with real-time numeric stream data.

2	**Decentralized Edge-Agentic Processing of Kafka Telemetry for Closed-Loop Crop Stress Remediation** **In:** ACM Transactions on Cyber-Physical Systems, 2025.	Deploys lightweight AI agents directly on edge nodes consuming local Kafka partition streams to orchestrate closed-loop irrigation and nutrient fertigation. Demonstrates that local edge-agent reasoning eliminates cloud round-trip latency, enabling real-time mitigation of acute water stress.	High cloud latency preventing sub-minute automated interventions for severe crop stress, network instability in remote agricultural zones, and heavy edge-to-cloud bandwidth overhead.
#	**Authors, paper, venue and DOI**	**Key contribution**	**Research gap**
3	**Agentic Guardrails and Event-Driven Stream Verification for Autonomous Agricultural Robotics** **In:** Journal of Field Robotics, 2025.	Develops a verification framework using supervisory AI agents that monitor Kafka control streams emitted by autonomous field robots inspecting crop health. Demonstrates that agentic guardrails can interdict unsafe autonomous actions (e.g., erroneous chemical dosing or collision paths) in real time while allowing adaptive pathing.	Safety risks and unpredictability of fully unconstrained autonomous agentic actions in physical field operations, and latency bottlenecks in traditional centralized safety verification loops.
4	**Autonomous Agentic Orchestration over Streaming Kafka Topics for Multi-Crop Disease Diagnosis** **In:** Computers and Electronics in Agriculture, 2025.	Deploys an ensemble of autonomous multi-agent workers directly subscribed to Kafka event streams to dynamically diagnose multi-crop pathological stress. Demonstrates how agentic workflows can execute real-time tool calling (retrieving weather forecasts, invoking vision models) directly triggered by incoming sensor telemetry.	Static stream processing rules failing to adapt to multi-factor environmental context, high latency in manual expert diagnosis, and lack of autonomous tool-calling integration within real-time streaming pipelines.
5	**Multi-Agent Collaborative Reasoning Framework for Distributed Field Telemetry and Crop Health Management** **In:** IEEE Internet of Things Journal, 2025.	Integrates event-driven Apache Kafka brokers with a specialized Multi-Agent System (MAS) where individual agents specialize in soil hydrology, crop pathology, and microclimate forecasting	Monolithic rule-based alert systems producing high false alarm rates during transient environmental spikes, and isolated domain models failing to synthesize cross-disciplinary agronomic data in real time.

6	**Real-Time Event-Driven Agentic AI for Dynamic Pest Risk Assessment and Site-Specific Spraying** **In:** IEEE Access, 2026.	Combines real-time Kafka event streams with reactive AI agents to continuously update spatial pest dispersion risks across large-scale farm clusters.	Inability of scheduled batch processing to react to rapid pest migration, high latency in generating actionable VRA maps from raw sensor streams, and blanket chemical usage.
7	**Self-Healing Agentic Workflows for Fault-Tolerant Stream Analytics in Precision Agriculture** **In:** Future Generation Computer Systems, 2025.	Implements self-correcting agentic monitoring agents on Kafka stream consumers to detect, diagnose, and impute corrupted or missing crop sensor data in real time.	Unreliable stream analytics caused by sensor drift, hardware degradation, or packet drops in harsh field environments, and rigid imputation heuristics that fail under dynamic weather shifts.
8	**Real-time Analytics in Edge-Cloud Systems** **In:** ACM Transactions on Cyber-Physical Systems, 2025.	Develops a decoupled inference architecture embedding lightweight ONNX runtimes directly into Apache Flink streaming workers for PyTorch models. Reduces CPU utilization by 35% compared to inline Python evaluation during continuous stream scoring.	Severe resource contention when running deep learning anomaly detection models directly on stream processors, and pipeline disruption during stateful execution graph restarts for model updates.
9	**Scalable Stream Processing for Agriculture** **In:** Computers and Electronics in Agriculture, 2026.	Combines replicated Kafka partition strategies with Flink state checkpointing deployed via Docker and Kubernetes across a simulated 500-node agricultural grid. Achieves zero data loss and rapid failover (<2 seconds recovery time) during node crash tests.	Lack of fault tolerance mechanisms during broker node failures in distributed farm monitoring systems, and high WAN network bandwidth consumption during state checkpointing.
10	**Online Anomaly Detection in Telemetry Streams** **In:** Data Mining and Knowledge Discovery, 2025.	Implements an online incremental isolation Forest model with dynamic sliding window updates in Python and Kafka Streams. Achieves a 94% F1-score while continuously adapting to ambient seasonal drift in microclimate and soil telemetry streams	High false-positive rates in dynamic agricultural telemetry due to static thresholding, and unmanaged memory growth over prolonged runtime without aggressive state cleanup mechanisms.

11	**Streaming Time Series Anomaly Detection In:** IEEE TKDE, 2026.	Combines Graph Neural Networks (GNN) with continuous sliding window attention mechanisms using PyTorch Geometric, Apache Flink, and Kafka. Outperforms baseline ARIMA and LSTM models by 18% in precision across multi-plot regional sensor networks.	Inability of traditional models to detect multi-variable spatial-temporal anomalies across adjacent land plots simultaneously, and high computational complexity limiting edge execution.
12	**Data lakes versus data warehouses In:** Journal of Electrical Systems and Information Technology, 2025.	Proposes a multizone data lake architecture (Raw, Trusted, Access) integrating real-time streaming tools such as Hadoop, Spark, Kafka, and Drill.	Rigid schema-on-write constraints of traditional data warehouses that fail to adapt to evolving IoT data streams, alongside complex governance and time-series security in data lakes.
13	**FIWARE and Apache KAFKA for precision agriculture robots In:** Agriculture, 2023.	Implements a distributed event bus pairing the FIWARE Context Broker with Apache Kafka and ROS for real-time agricultural robot telemetry. Achieves low end-to-end communication latency required for live robotic navigation and field map execution.	High latency in cloud-based robotic control loop communications and high edge hardware footprint requirements to support FIWARE Orion instances
14	**Big Data and Stream Processing for Smart Precision Agriculture In:** WSEAS Transactions, 2025.	Deploys a hybrid stream/batch architecture combining a Kafka buffer, Apache Flink processing engine, and HDFS cold storage for multispectral imagery and IoT streams. Achieves sub-second processing latency for streaming spatial and numerical metrics	Failure of standard batch processing on HDFS to handle low-latency real-time telemetry inputs, and state management overhead in Flink when processing out-of-order sensor readings
15	**Low-latency streaming architecture for IoT In:** IEEE Access, 2024.	Designs a partitioned Apache Kafka ingestion layer coupled with Spark Streaming sliding window calculations for synthetic time-series streams. Demonstrates ultra-low end-to-end latency (<50 ms) while handling high message throughput exceeding 100,000 events/second.	High end-to-end ingestion latency during peak telemetry bursts in edge-to-cloud setups, and resource consumption spikes triggered during sliding window aggregation executions.

16	**Crop Water Stress Index and Spatial Machine Learning** **In:** Remote Sensing, 2025.	Integrates satellite imagery, agroclimatic data, and in-situ soil moisture sensors using deep learning sensor fusion in olive orchards. Enables spatially distributed, scalable water-stress monitoring and precision irrigation management	Difficulty in scalable water-stress monitoring across spatially distributed orchards, and complex cross-calibration requirements across heterogeneous sensor platforms.
17	**Deep Learning Sensor Fusion in Plant Water Stress Assessment** **In:** Sensors, 2026.	Introduces SummaCConv, a sentence-level NLI-based method for detecting inconsistencies in summaries by aligning granularity between NLI tasks and summarization. Achieves improved accuracy in faithfulness evaluation.	Addresses mismatch between sentence-level NLI training and document-level evaluation. Improves reliability of hallucination detection compared to earlier NLI-based approaches.
18	**Stream Processing of Distributed Storage of Manufacturing Data** **In:** Applied Sciences, 2025.	Designs an Apache Kafka and PostgreSQL streaming pipeline where edge servers act as producers and Python consumer groups process topic-specific streams.	Coupling of message transmission and storage causing data loss under heavy load, and general IoT architecture limitations needing adaptation for agricultural telemetry.
19	**Edge-Fog-Cloud Architecture for monitoring IoT and robotics data** **In:** Computers, 2026.	Implements an Edge-Fog-Cloud streaming architecture utilizing MQTT brokers, Apache Kafka, GeoFlink, PostGIS, Hadoop, and Spark. Delivers 10 Hz continuous transmission and monitoring for agricultural robotics telemetry and IoT numeric streams.	Ingestion bottlenecks when processing real-time spatial and numeric streams from field robots, and poor scalability in traditional relational data warehouses.
20	**Edge AI and IoT for Real-time Crop Disease Detection** **In:** IJRIAS, 2025.	Deploys lean deep learning models (MobileNet, EfficientNet, CNNs) on edge hardware (Raspberry Pi, Jetson Nano) over LoRaWAN for real-time crop disease detection. Reduces energy consumption and communication bandwidth compared to cloud-reliant solutions.	High latency and data bandwidth sparsity in cloud-reliant crop anomaly detection, edge-model scalability constraints, and communication security issues.

21	**AI and IoT-integrated smart agriculture: Predictive crop health monitoring** **In:** IJRAR, 2025.	Evaluates hybrid deep learning models processing temporal soil sensor data (N, P, K, pH, EC, OC) and aerial imagery for tropical black pepper plantations. Utilizes SHAP for model interpretability, identifying pH, nitrogen, and leaf texture variance as top health predictors	Lack of crop-specific multimodal contextual calibration for perennial crops, along with unaddressed real-time latency challenges under actual field conditions.
22	**Smart Greenhouses in the Era of IoT and AI** **In:** Agriculture, 2026.	Synthesizes over 130 dataset studies (2020–2025) using multi-sensor ARX, ARMAX, and deep learning fusion for microclimate control. Achieves high air-temperature prediction accuracy ($R^2 = 0.965$) and improves crop stress diagnosis.	High vulnerability of crops to extreme climate and microclimate abiotic stressors, and seasonal drift risks caused by evaluations limited to single growing cycles
23	**Object detection with attributable tagging task** **In:** Frontiers in Artificial Intelligence, 2021.	Applies deep learning segmentation with Mask R-CNN and attributable tagging to detect crop anomalies and fruit in complex agricultural imaging datasets	Inability of standard object detection models to generalize across complex agricultural environments, and over-reliance on visual data without numerical stream integration.
24	**Digital agriculture in the transition from Agriculture 4.0 to Agriculture 5.0** **In:** Frontiers, 2026.	Conducts a comprehensive literature review (1981–2025) on AI, IoT, Blockchain, and WSN technologies in digital agriculture. Validates combining weather data with soil sensors for holistic decision-making and automated farm responses.	Inability to predict pest outbreaks proactively due to isolated data silos, high financial barriers to full farm automation, and operational complexity
25	**Smart irrigation monitoring and control strategies** **In:** Agricultural Water Management, 2022.	Develops edge computing and machine learning workflows using Random Forest Regression (RFR) algorithms for real-time smart irrigation monitoring. Provides a fundamental framework for crop water stress index modeling.	Fragmented workflows when transitioning raw sensor data into quantitative decision-making, high reliance on edge computing resources, and site-specific calibration constraints

26	**Adaptive Internet of Things (IoT) Implementation Framework** **In:** MAPACE, 2026.	Introduces a 7-layer adaptive IoT framework tailored for margin lands to accommodate context-based decision-making mechanisms across sensor networks. Delivers a scalable reference model for sustainable agricultural IoT deployments.	Lack of a holistic framework addressing cross-layer interactions in margin lands and limited empirical validation regarding long-term user adoption.
27	**Advances in Precision Agriculture** **In:** Archives of Current Research International, 2025.	Quantifies baseline efficiency gains, demonstrating 20–50% water savings and 15–30% fertilizer reduction. Validates site-specific application over uniform blanket chemical treatments.	Environmental pollution and suboptimal crop performance caused by blanket chemical applications, high initial investment costs, and technical data management hurdles.

28	**Smart IoT-driven precision agriculture In:** PLOS One, 2025.	Integrates IoT sensors (pH, soil moisture, multi-spectral NIR/red band imagery) using fuzzy logic and K-means clustering to optimize irrigation and soil nutrient management. Demonstrates open-source baseline sensor integration and automated power conservation, significantly reducing water waste compared to manual irrigation.	Inefficient water usage, lack of real-time soil nutrient mapping, and spatial clustering limits requiring improved feature selection techniques.
29	**From Precision Agriculture to Intelligent Agricultural Ecosystems** **In:** Agronomy, 2026.	Establishes a systematic mapping review connecting precision agriculture to intelligent ecosystems via predictive analytics. Demonstrates that coupling IoT stream data with big data frameworks (Apache Kafka, Storm, MongoDB, DBSCAN) enhances yield efficiency. Connects stream processing to agricultural stream enrichment and anomaly detection.	Fragmented applications of machine learning and big data in agriculture, along with the lack of empirical latency benchmarking for Kafka/Storm streaming stacks.
30	**Front. Sustain. Food Syst. (Ag 5.0 Transition Framework)** **In:** Frontiers in Sustainable Food Systems, 2026.	Integrates IoT, AI, WSNs, and robotics to define future smart farming paradigms. Provides theoretical grounding for predictive stream analytics in smart agricultural ecosystems.	Terminological ambiguity and structural constraints in transitioning to fully automated Agriculture 5.0, as well as a lack of empirical validation for conceptual Ag 5.0 models.

2. Research Gap and Positioning

The 30 papers show that Kafka and Flink can handle fast sensor streams and that agentic AI can interpret them. Few combine both with fault tolerance, grounded recommendations and human approval. Our Project addresses these gaps as follows:

- **Latency (cloud-reliant pipelines):** Kafka partitions and PySpark Structured Streaming process sensor events continuously rather than in batches. The pipeline runs locally (Docker Compose, with Ollama as an option), and we measure end-to-end throughput and P99 latency.
- **Static methods and false alerts:** instead of fixed thresholds, we use windowed detection with EWMA, rolling z-score and Isolation Forest. These are calibrated to the baseline, because natural flatlines and jumps in our data are not faults. We compare them against rule-threshold, statistical and zero-shot LLM baselines.
- **Data fusion and explainability:** a triage agent receives only detected events and returns structured output: stress, fault or noise, with confidence and evidence. A RAG advisor fuses these with text knowledge from FAO and ICAR sources, so farmers get explained, actionable advice.
- **Fault tolerance:** replicated Kafka topics and Spark checkpoints support recovery. We test this with chaos experiments (killing the broker, a Spark worker or the agent, and blocking the LLM API) and measure mean time to recovery, message loss and duplicate processing.
- **Safety and verification:** guardrails enforce strict schema validation and citation checks, and block chemical-dosage advice. A human-in-the-loop approval step (Telegram bot and Grafana dashboard) must clear every recommendation before anyone acts on it.
- **Scalability and validation:** we scale from 100 to 1,000 virtual fields and from 1 to 3 to 6 Kafka partitions. A benchmark of about 60 labelled scenarios gives reproducible metrics: macro-F1, alert reduction, grounding % and unsafe-advice rate.